



2013 Trial Examination

FORM VI

MATHEMATICS EXTENSION 2

Thursday 1st August 2013

General Instructions

- Reading time — 5 minutes
- Writing time — 3 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 100 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 90 Marks

- Questions 11–16 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Eleven.
- Write your candidate number on this question paper and submit it with your answers.

Checklist

- SGS booklets — 6 per boy
- Multiple choice answer sheet
- Candidature — 68 boys

Examiner
DS/REP

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

Which function is a primitive of $\frac{2x}{2x-1}$?

- (A) $x + \ln(2x - 1)$
- (B) $\ln(2x - 1)$
- (C) $x + \frac{1}{2} \ln(2x - 1)$
- (D) $\frac{1}{2} \ln(2x - 1)$

QUESTION TWO

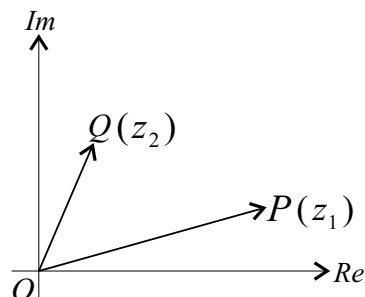
Which expression is a correct factorisation of $z^3 - i$?

- (A) $(z - i)(z^2 + iz + 1)$
- (B) $(z + i)(z^2 - iz - 1)$
- (C) $(z + i)(z - i)^2$
- (D) $(z + i)^3$

QUESTION THREE

If $f(x)$ is an odd function, which statement is true?

- (A) $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$
- (B) $\int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx$
- (C) $\int_{-2a}^a f(x) dx = \int_{-a}^{2a} f(x) dx$
- (D) $\int_{-a}^{2a} f(x) dx = \int_a^{2a} f(x) dx$

QUESTION FOUR

The points P and Q in the first quadrant represent the complex numbers z_1 and z_2 respectively, as shown in the diagram above. Which statement about the complex number $z_2 - z_1$ is true?

- (A) It is represented by the vector QP .
- (B) Its principal argument lies between $\frac{\pi}{2}$ and π .
- (C) Its real part is positive.
- (D) Its modulus is greater than $|z_1 + z_2|$.

QUESTION FIVE

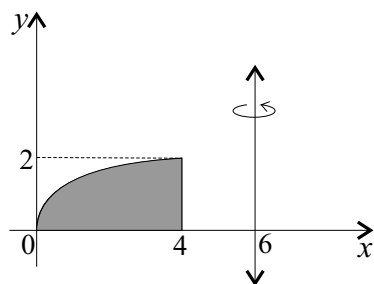
An ellipse has foci at $(-6, 0)$ and $(6, 0)$, and its directrices have equations $x = -8$ and $x = 8$. What is the eccentricity of the ellipse?

- (A) $\frac{1}{2}\sqrt{3}$
- (B) $\frac{1}{3}\sqrt{3}$
- (C) $\frac{2}{3}\sqrt{3}$
- (D) $3\sqrt{3}$

QUESTION SIX

The polynomial $P(x) = x^3 + 3x^2 - 24x + 28$ has a double zero. What is the value of the double zero?

- (A) -7
- (B) -4
- (C) 4
- (D) 2

QUESTION SEVEN

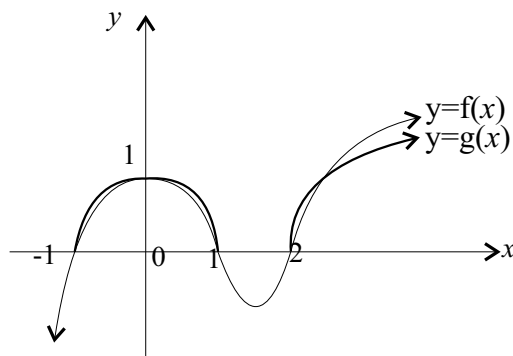
The diagram above shows the region bounded by the curve $y = \sqrt{x}$ and the x -axis, from $x = 0$ to $x = 4$. The region is rotated about the line $x = 6$ to form a solid of revolution. Which integral gives the volume of the solid?

- (A) $\int_0^2 \pi(4 - y^2)(8 - y^2) dy$
- (B) $\int_0^2 4\pi(5 - y^2) dy$
- (C) $\int_0^2 \pi(4 - y^2)(6 - y^2) dy$
- (D) $\int_0^2 \pi(2 - y^2)(6 - y^2) dy$

QUESTION EIGHT

The curve defined by the equation $x^2 - xy + 2y^2 = 4$ passes through the point $P(1, -1)$. What is the gradient of the tangent to the curve at P ?

- (A) $\frac{1}{2}$ (B) $-\frac{1}{5}$ (C) $\frac{3}{5}$ (D) $\frac{3}{4}$

QUESTION NINE

In the diagram above, the relationship between the functions $f(x)$ and $g(x)$ could be represented by:

(A) $g(x) = (f(x))^2$

(B) $g(x) = \log_e f(x)$

(C) $g(x) = \sqrt{f(x)}$

(D) $g(x) = |f(x)|$

QUESTION TEN

Without attempting to evaluate the integrals, determine which of the following inequalities is FALSE:

(A) $\int_1^2 \frac{1}{1+x} dx < \int_1^2 \frac{1}{x} dx$

(B) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x}{x} dx < \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{x} dx$

(C) $\int_0^{\frac{\pi}{4}} \tan^2 x dx < \int_0^{\frac{\pi}{4}} \tan^3 x dx$

(D) $\int_1^2 e^{-x^2} dx < \int_0^1 e^{-x^2} dx$

_____ End of Section I _____

Exam continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

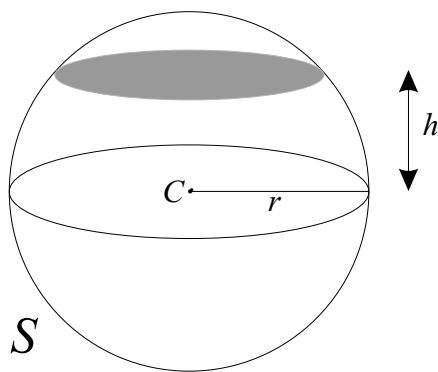
Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet.	Marks
(a) Solve the quadratic equation $4z^2 + 4z + 5 = 0$.	2
(b) Find the real values of x and y for which $\frac{x}{i} - \frac{y}{1+i} = -1 - 3i$.	2
(c) Find $\int \frac{3x}{2x^2 - 5x + 2} dx$.	3
(d) Find $\int \frac{1}{\sqrt{x^2 + 6x + 34}} dx$.	2
(e) Evaluate $\int_0^{\frac{\pi}{3}} \tan^4 x \, dx$.	3
(f) (i) Use the substitution $u = a - x$ to prove that $\int_0^a f(x) \, dx = \int_0^a f(a - x) \, dx$.	1
(ii) Hence find the value of $\int_0^1 x(1 - x)^7 \, dx$.	2

QUESTION TWELVE (15 marks) Use a separate writing booklet.

Marks

- (a) Shade the region in the Argand diagram where $|z + 3i| > 2|z|$. 3
- (b) (i) Express $-2 + 2i$ in modulus-argument form. 1
- (ii) Simplify $(-2 + 2i)^{8k}$, where k is an integer. 2
- (c) A complex number z satisfies $\arg(z - 1) = \frac{\pi}{6}$.
- (i) Sketch the locus of z . 1
- (ii) Show that $|z - 5| \geq 2$. 1
- (d)

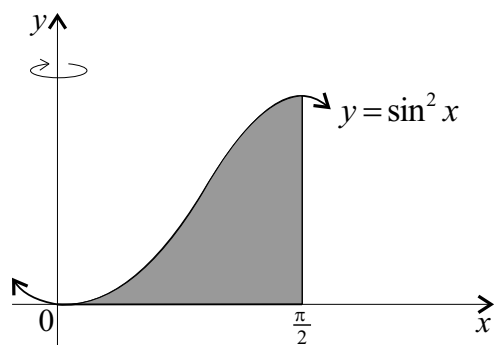


In the diagram above, S is a sphere of radius r . The point C is the centre of the sphere. A typical horizontal cross-section h units above C is shown.

- (i) Find the area of this cross-section as a function of h . 1
- (ii) Hence prove that the volume of S is $\frac{4}{3}\pi r^3$. 2
- (e) The polynomial $P(x) = x^4 - 4x^3 + 10x^2 - 12x - 40$ has zeroes α, β, γ and δ .
- (i) Find a polynomial with zeroes $\alpha - 1, \beta - 1, \gamma - 1$ and $\delta - 1$. 2
- (ii) Hence find the zeroes of $P(x)$. 2

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) (i) Write down the equation of a line with gradient m that passes through the point $T(-4c, 2c)$. 1
- (ii) Solve the equation in part (i) simultaneously with the equation $xy = c^2$ to obtain a quadratic equation in x . 2
- (iii) Hence find the gradients of the two tangents to the rectangular hyperbola $xy = c^2$ that pass through the point $T(-4c, 2c)$. 2
- (b) Consider the polynomial $P(z) = z^4 + (1 - 2i)z^2 - 2i$.
- (i) Show that $P(i) = 0$. 1
- (ii) Explain why $P(-i)$ must also be zero. 1
- (iii) Suppose that the other two zeroes of $P(z)$ are w and $-w$.
Use the product of the zeroes to find w . 3
- (c) (i) Find $\int x \cos 2x \, dx$. 2
- (ii) 3



The diagram above shows the region bounded by the curve $y = \sin^2 x$, the x -axis and the line $x = \frac{\pi}{2}$. The region is rotated about the y -axis through 360° . Use the method of cylindrical shells to find the exact volume of the solid formed.

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) An object of mass 2 kg is projected vertically upwards at 20 m/s and experiences air resistance of magnitude $\frac{1}{10}v^2$ Newtons, where v is the speed of the object after t seconds. Take $g = 10 \text{ m/s}^2$.

(i) Show that the maximum height reached by the object is $10 \ln 3$ metres.

3

(ii) Find the speed of the object, correct to three significant figures, at the instant it reaches half its maximum height.

2

- (b) Let $I_n = \int_1^e (1 + \ln x)^n dx$, where $n \geq 0$.

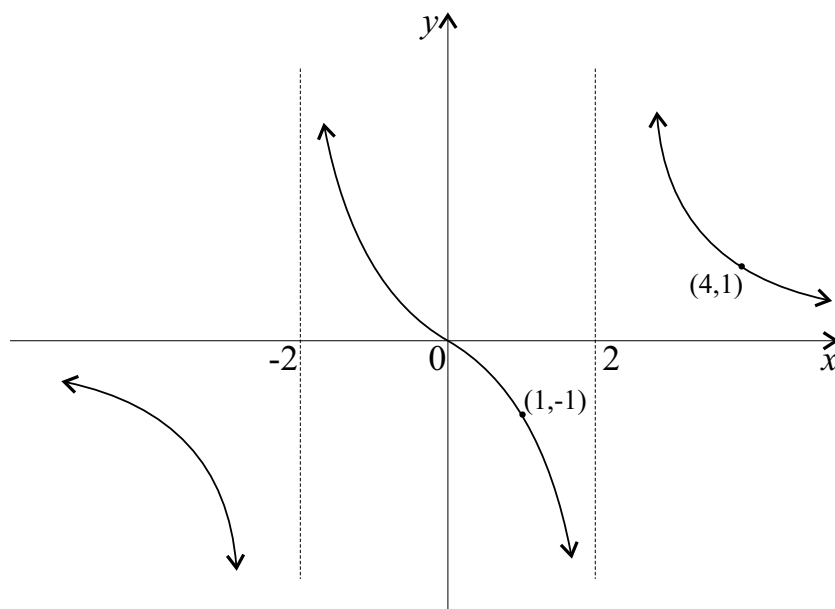
(i) Use integration by parts to show that $I_n = (2^n)e - 1 - nI_{n-1}$.

2

(ii) Hence find the exact value of $\int_1^e (2 + \ln x)(1 + \ln x)^2 dx$.

2

(c)



The diagram above shows the graph of the odd function $y = f(x)$, where

$$f(x) = \frac{3x}{x^2 - 4}.$$

Sketch the graphs of each of the following functions on large separate diagrams, showing the x -intercepts and asymptotes. You are NOT expected to find any stationary points.

(i) $y = \frac{1}{f(x)}$

2

(ii) $y = \log_e f(x)$

2

(iii) $y = x + f(x)$

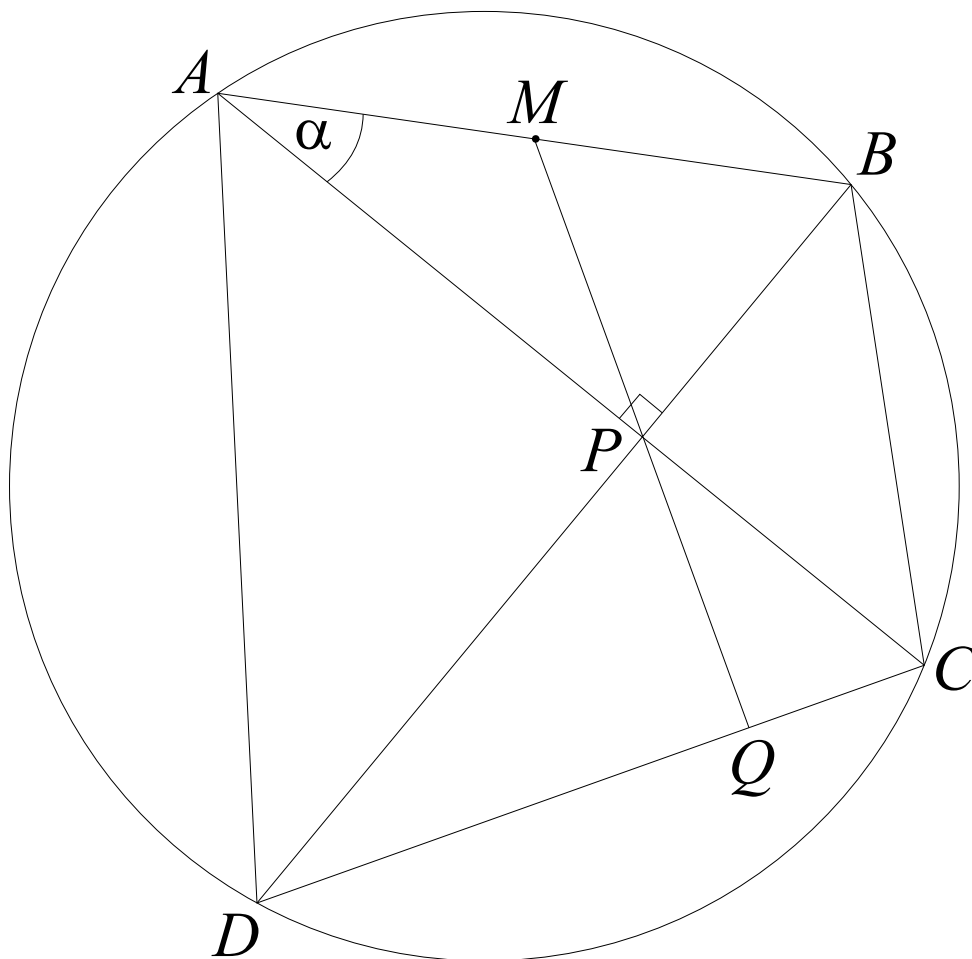
2

Exam continues overleaf ...

QUESTION FIFTEEN (15 marks) Use a separate writing booklet.

Marks

(a)



In the diagram above $ABCD$ is a cyclic quadrilateral whose diagonals are perpendicular and intersect at P . Let M be the midpoint of AB , and suppose that MP produced meets DC at Q . Let $\angle PAM = \alpha$.

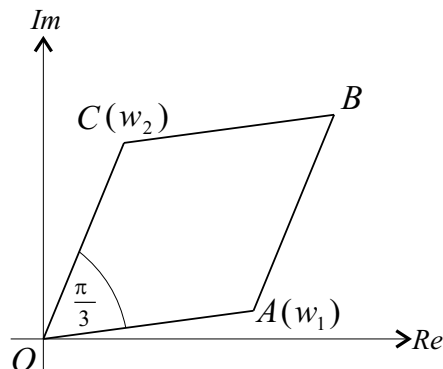
- (i) Explain why $AM = PM$.
- (ii) Prove that $MQ \perp DC$.

1

3

QUESTION FIFTEEN (Continued)

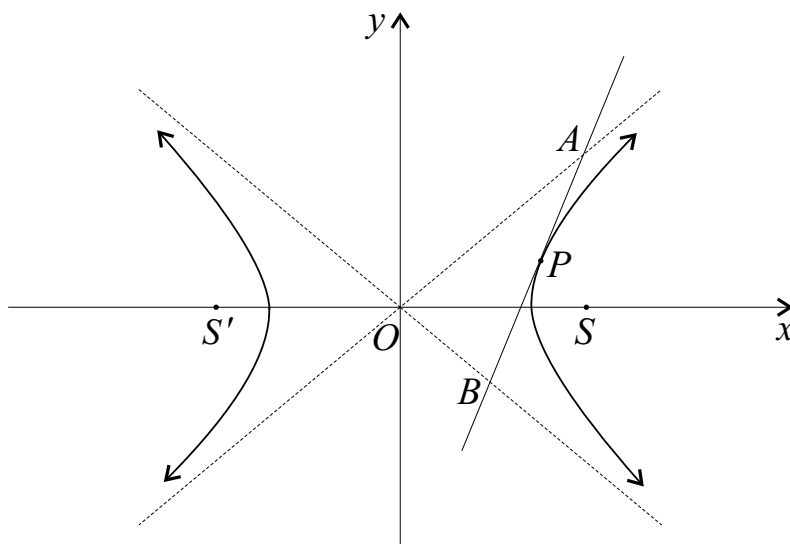
(b)



In the Argand diagram above, $OABC$ is a rhombus with $\angle COA = \frac{\pi}{3}$. The points A and C represent the complex numbers w_1 and w_2 respectively.

- (i) Explain why $w_2 = w_1 \operatorname{cis} \frac{\pi}{3}$. 1
- (ii) Write down, in terms of w_1 only, the complex numbers represented by the vectors OB and AC . 2
- (iii) By considering $i(w_1 + w_2)$, show that the diagonals OB and AC of the rhombus are perpendicular. 2

(c)



The diagram above shows the tangent at a point $P(a \sec \theta, b \tan \theta)$ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meeting the asymptotes of the hyperbola at A and B . The points S and S' are the foci of the hyperbola.

- (i) Show that the tangent at P has equation $bx \sec \theta - ay \tan \theta = ab$. 1
- (ii) Show that $OA \times OB = a^2 + b^2$, where O is the origin. 2
- (iii) Extend AO to A' so that $OA' = OB$ and extend BO to B' so that $OB' = OA$. Explain why the points A, B, A' and B' are concyclic. 1
- (iv) Hence show that the points A, S, B and S' are concyclic. 2

Exam continues overleaf ...

QUESTION SIXTEEN (15 marks) Use a separate writing booklet.**Marks**(a) Let $z = \cos \theta + i \sin \theta$ and suppose that n is a positive integer.(i) Show that $z^n + z^{-n} = 2 \cos n\theta$. 1(ii) Hence use the identity $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$ to show that 2

$$(z^{2n} + z^{2n-2} + z^{2n-4} + \cdots + z^{-2n}) \sin \theta = \sin(2n+1)\theta.$$

(iii) Use part (ii) and the identity $\cos 3A = 4 \cos^3 A - 3 \cos A$ to deduce that 1

$$8 \cos^3 2\theta + 4 \cos^2 2\theta - 4 \cos 2\theta - 1 = \frac{\sin 7\theta}{\sin \theta}.$$

(iv) Hence show that $\cos \frac{2\pi}{7}$ is a root of the equation $8x^3 + 4x^2 - 4x - 1 = 0$. 1(b) (i) Use a suitable double angle formula to show that $\tan \frac{\pi}{8} = \sqrt{2} - 1$. 1(ii) Find $\cos 4\theta$ in terms of $\cos \theta$. 1(iii) Let $I = \int_{-1}^1 \frac{1}{\sqrt{1+x} + \sqrt{1-x} + 2} dx$.
 (α) Show, by using the substitution $x = \sin 4\theta$, that 2

$$I = \int_0^{\frac{\pi}{8}} \frac{2 \cos 4\theta}{\cos^2 \theta} d\theta.$$

 (β) Hence find the exact value of I . 2
(c) (i) Explain why 1

$$\left(1 - \frac{1}{2}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{5} - \frac{1}{6}\right) + \cdots + \left(\frac{1}{2k-1} - \frac{1}{2k}\right) > \frac{k}{(2k+1)(2k+2)}.$$

(ii) Prove by mathematical induction, or otherwise, that for all $n \geq 2$, 3

$$n \left(1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n-1}\right) > (n+1) \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \cdots + \frac{1}{2n}\right).$$

 End of Section II

END OF EXAMINATION

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

CANDIDATE NUMBER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

Multiple Choice

$$(1) \int \frac{2x}{2x-1} dx = \int \left(1 + \frac{1}{2x-1}\right) dx$$

$$= x + \frac{1}{2} \ln(2x-1) + c$$

(C)

$$(2) z^3 - i = z^3 + i^3$$

$$= (z+i)(z^2 - iz - 1)$$

(B)

$$(3) \int_{-a}^{2a} f(x) dx = \int_{-a}^a f(x) dx + \int_a^{2a} f(x) dx$$

$$= 0 + \int_a^{2a} f(x) dx$$

(D)

$$(4) \frac{\pi}{2} < \arg(z_2 - z_1) < \pi$$

(B)

$$(5) ae = 6 \text{ and } \frac{a}{e} = 8$$

(A)

$$\text{So } 8e^2 = 6$$

$$e = \frac{\sqrt{3}}{2}$$

$$(6) P'(x) = 3(x^2 + 2x - 8)$$

$$= 3(x+4)(x-2)$$

$$P(-4) \neq 0$$

$$P(2) = 0$$

(D)

$$(7) A(y) = \pi(R-r)(R+r), \text{ where } \begin{cases} R = 6-x \\ r = 2 \end{cases}$$

$$= \pi(4-x)(8-x)$$

$$= \pi(4-y^2)(8-y^2)$$

(A)

$$(8) 2x - y - x \cdot \frac{dy}{dx} + 4y \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{y-2x}{4y-x}, m = \frac{-1-2}{-4-1}$$

(C)

(9) (C)

(10)

(C)

Written Response

$$(11)(a) \quad 4z^2 + 4z + 5 = 0$$

$$z = \frac{-4 \pm 8i}{8}$$

$$= -\frac{1}{2} \pm i$$

$$\Delta = 16 - 80$$

$$= -64$$

$$(b) \quad \frac{x}{i} - \frac{y}{1+i} = -1-3i$$

$$-ix - \frac{y(1-i)}{2} = -1-3i$$

$$-y + i(y-2x) = -2-6i$$

$$y=2 \text{ and } x=4$$

$$(c) \quad \int \frac{3x}{2x^2-5x+2} dx = \int \left(\frac{2}{x-2} - \frac{1}{2x-1} \right) dx$$

$$= 2 \ln|x-2| - \frac{1}{2} \ln|2x-1| + c$$

$$(d) \quad \int \frac{1}{\sqrt{x^2+6x+34}} dx = \int \frac{1}{\sqrt{(x+3)^2+25}} dx$$

$$= \ln \left| (x+3) + \sqrt{x^2+6x+34} \right| + c$$

$$(e) \quad \int_0^{\frac{\pi}{3}} \tan^4 x dx = \int_0^{\frac{\pi}{3}} \tan^2 x (\sec^2 x - 1) dx$$

$$= \int_0^{\frac{\pi}{3}} \tan^2 x \sec^2 x dx - \int_0^{\frac{\pi}{3}} (\sec^2 x - 1) dx$$

$$= \left[\frac{1}{3} \tan^3 x \right]_0^{\frac{\pi}{3}} - \left[\tan x - x \right]_0^{\frac{\pi}{3}}$$

$$= \frac{1}{3} \cdot 3\sqrt{3} - \left(\sqrt{3} - \frac{\pi}{3} \right)$$

$$= \frac{\pi}{3}$$

$$(f) \quad (i) \int_0^a f(x) dx$$

$$= \int_a^0 f(a-u) \cdot -du$$

$$= \int_0^a f(a-u) du$$

$$= \int_0^a f(a-x) dx$$

(dummy variable)

$$u = a - x$$

$$\text{So } dx = -du$$

x	0	a
u	a	0

$$(ii) \quad \int_0^1 x(1-x)^7 dx$$

$$= \int_0^1 (1-x)x^7 dx \quad \text{(using part (i))}$$

$$= \int_0^1 (x^7 - x^8) dx$$

$$= \left[\frac{x^8}{8} - \frac{x^9}{9} \right]_0^1$$

$$= \frac{1}{8} - \frac{1}{9}$$

$$= \frac{1}{72}$$

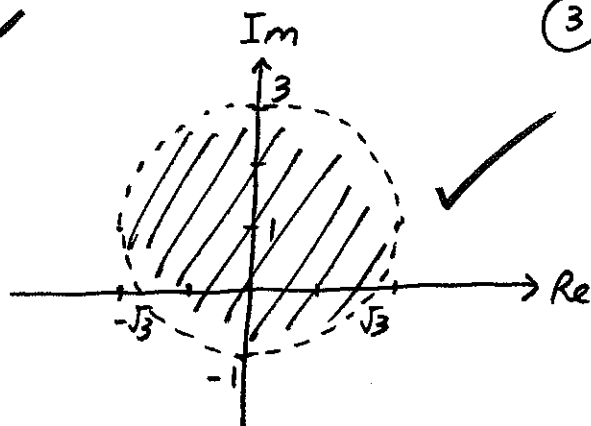
(12)(a) Let $z = x + iy$.

Then $x^2 + (y+3)^2 > 4(x^2 + y^2)$ ✓

$6y + 9 > 3x^2 + 3y^2$

$x^2 + y^2 - 2y < 3$ ✓

$x^2 + (y-1)^2 < 4$ ✓



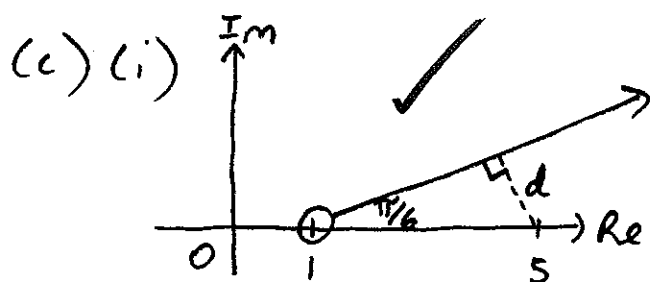
(b)(i) $-2 + 2i = 2\sqrt{2} \operatorname{cis} \frac{3\pi}{4}$ ✓

(ii) $(-2 + 2i)^{8k} = (2\sqrt{2})^{8k} \operatorname{cis} \left(\frac{3\pi}{4} \cdot 8k \right)$ ✓

$= 2^{8k} \cdot 2^{4k} \operatorname{cis}(6\pi k)$

$= 2^{12k} \operatorname{cis} 6\pi k$

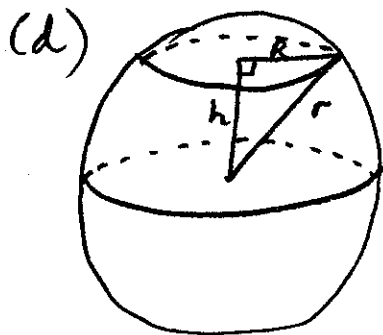
$= 2^{12k}, k \in \mathbb{Z}$ ✓



(ii) Let d be the minimum value of $|z-5|$.
(shown on diagram)

$d = 4 \sin \frac{\pi}{6}$
 $= 2$ ✓

So $|z-5| \geq 2$.



(i) $A(h) = \pi R^2$
 $= \pi(r^2 - h^2)$ ✓

(ii) $V = 2 \int_0^r \pi(r^2 - h^2) dh$ ✓
 $= 2\pi \left[r^2 h - \frac{1}{3} h^3 \right]_0^r$ ✓
 $= 2\pi \left(r^3 - \frac{1}{3} r^3 \right)$ ✓
 $= 2\pi \cdot \frac{2}{3} r^3$ ✓
 $= \frac{4}{3} \pi r^3$ ✓

(e)(i) $P(x+1) = (x+1)^4 - 4(x+1)^3 + 10(x+1)^2 - 12(x+1) - 40$ ✓
 $= x^4 + 4x^3 + 6x^2 + 4x + 1 - 4x^3 - 12x^2 - 12x - 4$
 $+ 10x^2 + 20x + 10 - 12x - 12 - 40$
 $= x^4 + 4x^2 - 45$ ✓

(ii) $P(x+1) = (x^2+9)(x^2-5)$ and so has zeroes $\pm 3i$ and $\pm \sqrt{5}$.
So $P(x)$ has zeroes $1 \pm 3i$ and $1 \pm \sqrt{5}$. ✓

$$(13)(a)(i) \quad y - 2c = m(x + 4c) \quad \checkmark$$

$$(ii) \quad y = mx + 2c(2m+1) \quad (1)$$

$$xy = c^2 \quad (2)$$

$$\text{Subst (1) into (2): } mx^2 + 2c(2m+1)x - c^2 = 0$$

(iii) Line (1) is a tangent to curve (2)

when $\Delta = 0$.

$$\text{So } 4c^2(2m+1)^2 + 4c^2m = 0 \quad \checkmark$$

$$4m^2 + 5m + 1 = 0$$

$$(4m+1)(m+1) = 0$$

$$m = -\frac{1}{4} \text{ or } -1 \quad \checkmark$$

$$(b) \quad P(z) = z^4 + (1-2i)z^2 - 2i$$

$$(i) \quad P(i) = i^4 + (1-2i)i^2 - 2i \\ = 1 - 1 + 2i - 2i \\ = 0 \quad \checkmark$$

(ii) $P(z)$ is even, since it only involves even powers of z . $\checkmark$

(iii) Product of zeroes is $-2i$. $\checkmark$

$$\text{So } i^2 w^2 = -2i$$

$$w^2 = 2i \quad \checkmark$$

$$\text{Let } w = x + iy.$$

$$\text{Then } x^2 - y^2 = 0 \text{ and } xy = 1.$$

$$\text{So } w = 1+i \text{ (or } -1-i). \quad \checkmark$$

$$(13)(c)(i) \int x \cos 2x \, dx$$

$$= uv - \int v u'$$

$$= \frac{1}{2} x \sin 2x - \int \frac{1}{2} \sin 2x \, dx$$

$$= \frac{1}{2} x \sin 2x + \frac{1}{4} \cos 2x + c$$

$$\text{Let } u = x. \quad (5)$$

$$\text{Then } u' = 1.$$

$$\text{Let } v' = \cos 2x.$$

$$\text{Then } v = \frac{1}{2} \sin 2x.$$

$$(ii) V = \int_0^{\frac{\pi}{2}} 2\pi r h \, dx, \text{ where } r = x \text{ and } h = y = \sin^2 x$$

$$= 2\pi \int_0^{\frac{\pi}{2}} x \sin^2 x \, dx$$

$$= 2\pi \int_0^{\frac{\pi}{2}} \frac{1}{2} x (1 - \cos 2x) \, dx$$

$$= 2\pi \int_0^{\frac{\pi}{2}} \frac{1}{2} x \, dx - 2\pi \int_0^{\frac{\pi}{2}} \frac{1}{2} x \cos 2x \, dx$$

$$= \pi \left[\frac{1}{2} x^2 \right]_0^{\frac{\pi}{2}} - \pi \left[\frac{1}{2} x \sin 2x + \frac{1}{4} \cos 2x \right]_0^{\frac{\pi}{2}}$$

$$= \pi \cdot \frac{\pi^2}{8} - \pi \left(-\frac{1}{4} - \frac{1}{4} \right)$$

$$= \pi \left(\frac{\pi^2}{8} + \frac{1}{2} \right)$$

$$= \frac{1}{8} \pi (\pi^2 + 4) \text{ units}^3$$

(14)(a)(i) $m\ddot{x} = -mg - \frac{v^2}{10}$

$$2\ddot{x} = -20 - \frac{v^2}{10}$$

$$\ddot{x} = \frac{-200 - v^2}{20}$$

$$v \cdot \frac{dv}{dx} = \frac{-200 - v^2}{20} \quad \checkmark$$

$$\frac{dx}{dv} = \frac{-20v}{200 + v^2}$$

$$x = -10 \int \frac{2v}{200 + v^2} dv$$

$$= -10 \ln(200 + v^2) + c \quad \checkmark$$

When $t=0$, $x=0$ and $v=20$.

$$\text{So } c = 10 \ln 600.$$

$$\text{So } x = 10 \ln \left(\frac{600}{200 + v^2} \right) \quad \checkmark$$

So when $v=0$, $x = 10 \ln 3$.

So the maximum height is $10 \ln 3$ metres.

(ii) When $x = 5 \ln 3$, $5 \ln 3 = 10 \ln \left(\frac{600}{200 + v^2} \right)$

$$e^{\frac{1}{2} \ln 3} = \frac{600}{200 + v^2} \quad \checkmark$$

$$\frac{200 + v^2}{600} = \frac{1}{\sqrt{3}}$$

$$200 + v^2 = 200\sqrt{3}$$

$$v^2 = 200(\sqrt{3} - 1) = 146.41 \dots$$

$$\text{So } v \doteq 12.1 \text{ ms}^{-1} \quad \checkmark$$

(b)(i) $I_n = [uv]_1^e - \int_1^e v u'$

$$= [x(1 + \ln x)^n]_1^e - n \int_1^e (1 + \ln x)^{n-1} dx$$

$$= e \cdot 2^n - 1 \cdot 1 - n \cdot I_{n-1}$$

$$= (2^n)e - 1 - n I_{n-1}$$

Let $u = (1 + \ln x)^n$.

Then $u' = \frac{n}{x} (1 + \ln x)^{n-1}$.

Let $v' = 1$.

Then $v = x$. $\checkmark$

$$(14)(b)(ii) \int_1^e (2 + \ln x)(1 + \ln x)^2 dx$$

$$= \int_1^e (1 + (1 + \ln x))(1 + \ln x)^2 dx \quad \checkmark$$

$$= I_2 + I_3$$

$$\text{Now, } I_0 = \int_1^e dx = e - 1$$

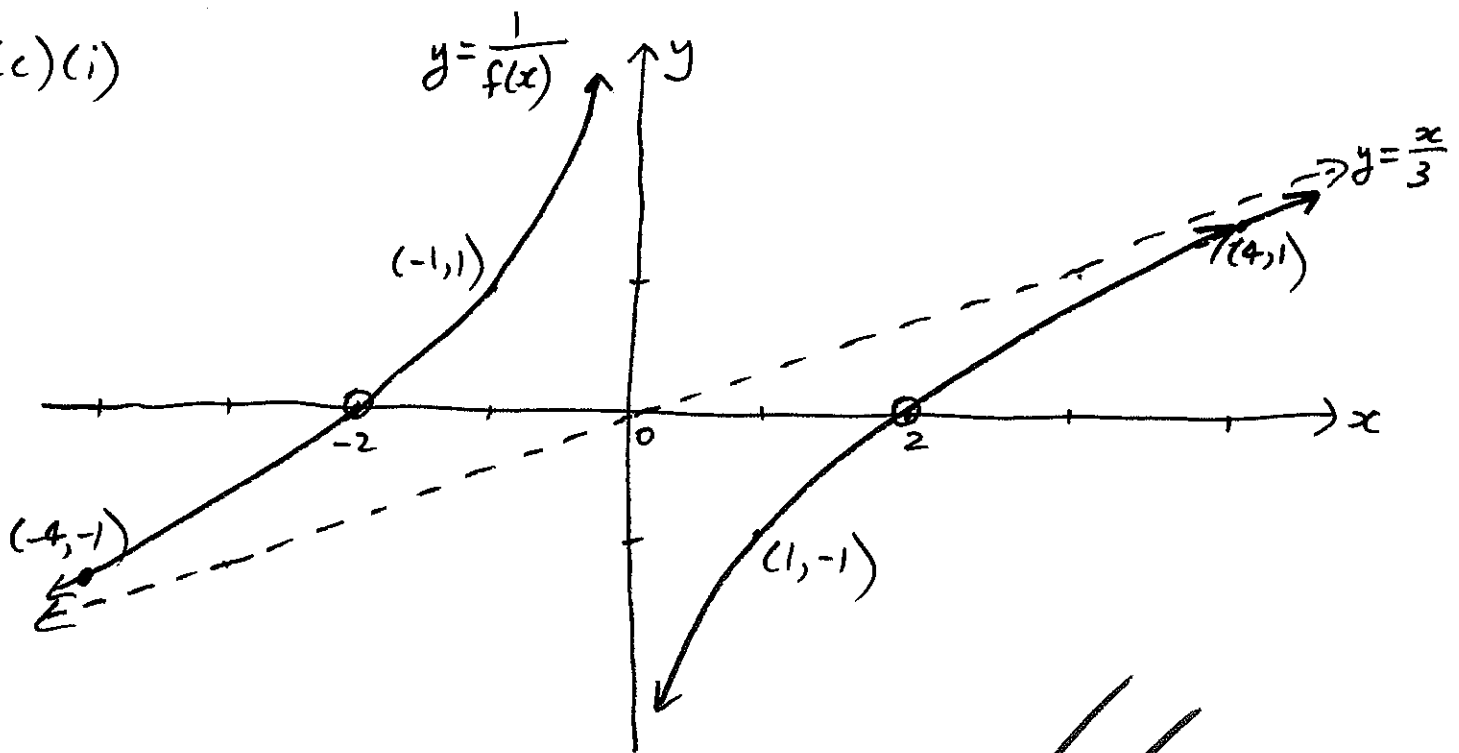
$$I_1 = 2e - 1 - (e - 1) = e$$

$$I_2 = 4e - 1 - 2e = 2e - 1$$

$$I_3 = 8e - 1 - 3(2e - 1) = 2e + 2.$$

$$\text{So } \int_1^e (2 + \ln x)(1 + \ln x)^2 dx = 4e + 1. \quad \checkmark$$

(c)(i)



Maximum of one mark
if either the open circles
or the oblique
asymptote are omitted.

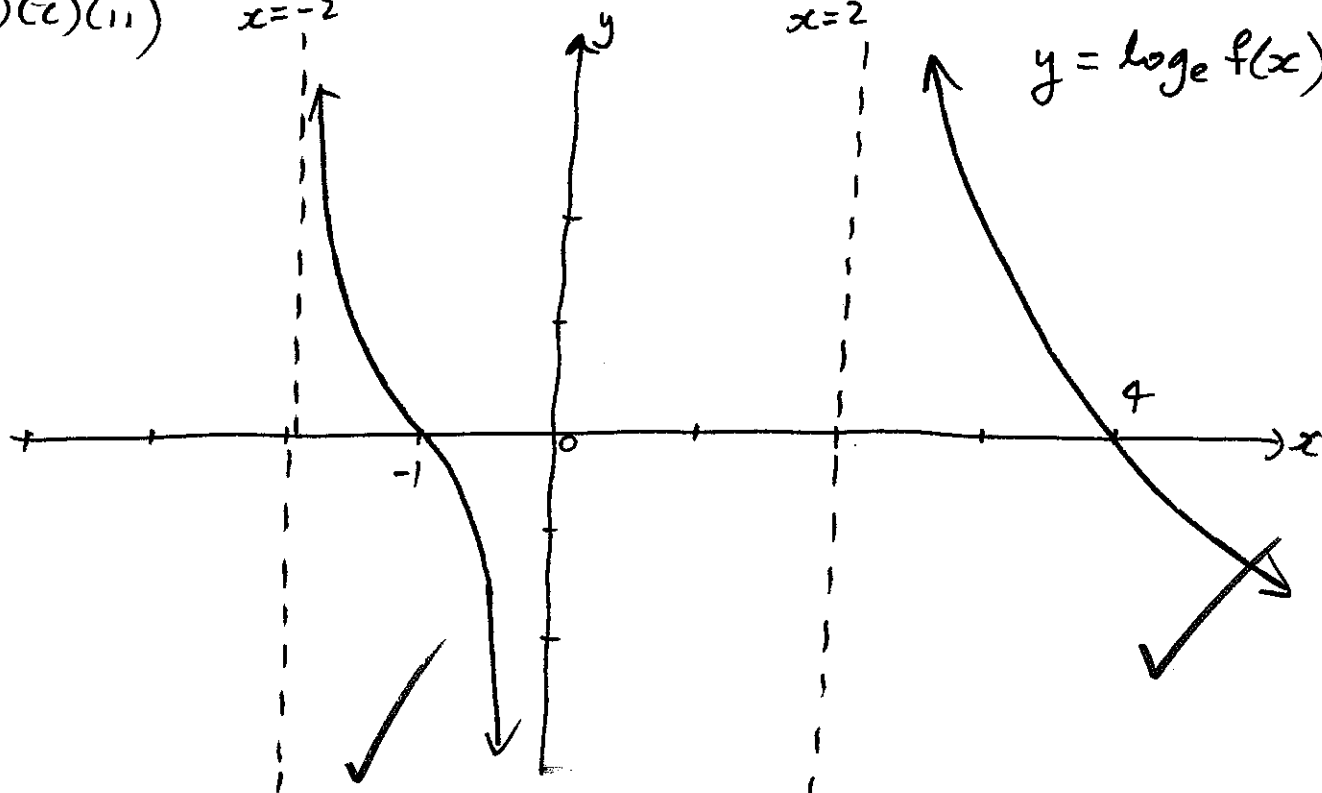
(14)(c)(ii)

$x = -2$

$x = 2$

$y = \log_e f(x)$

(8)

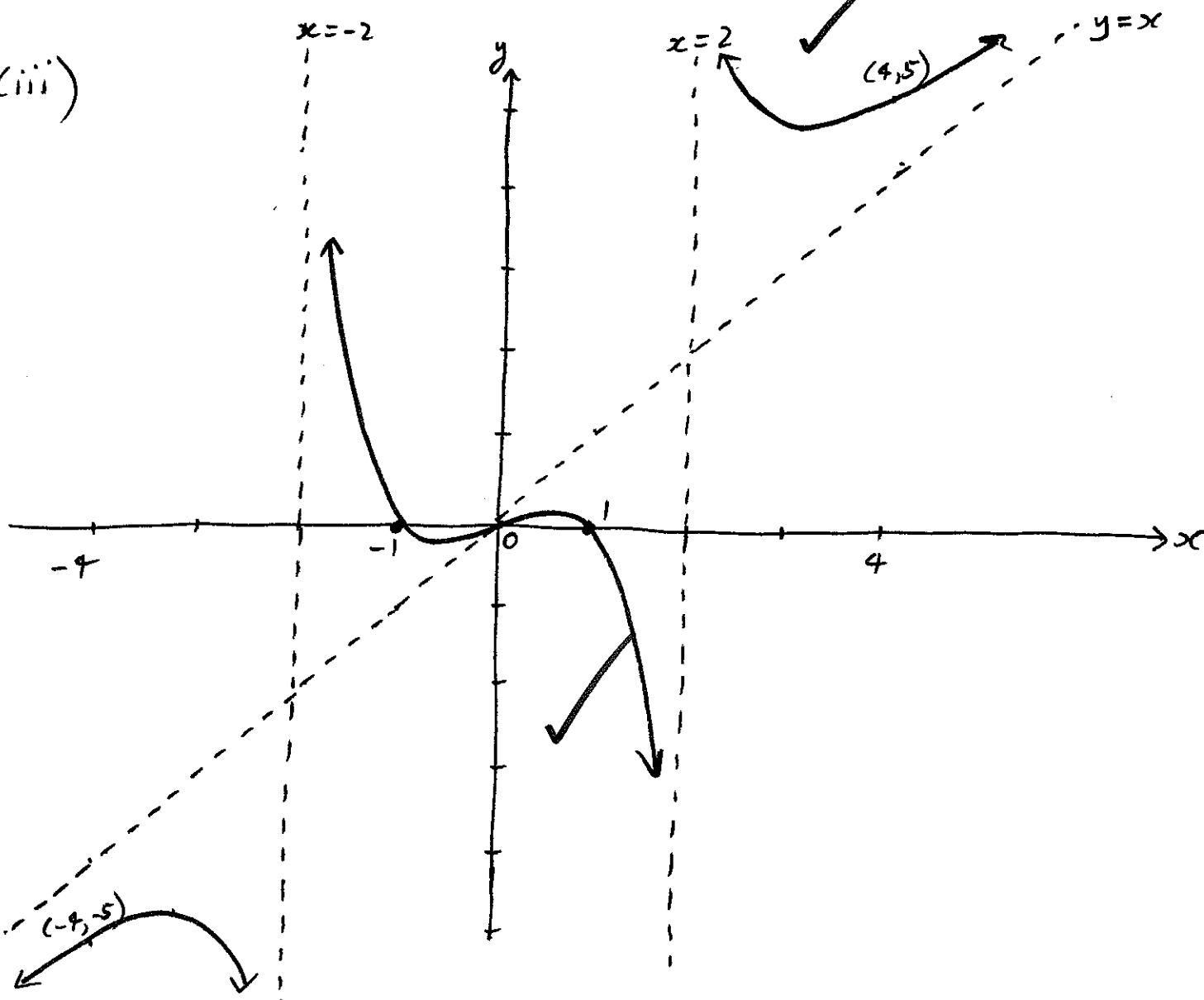


(iii)

$x = -2$

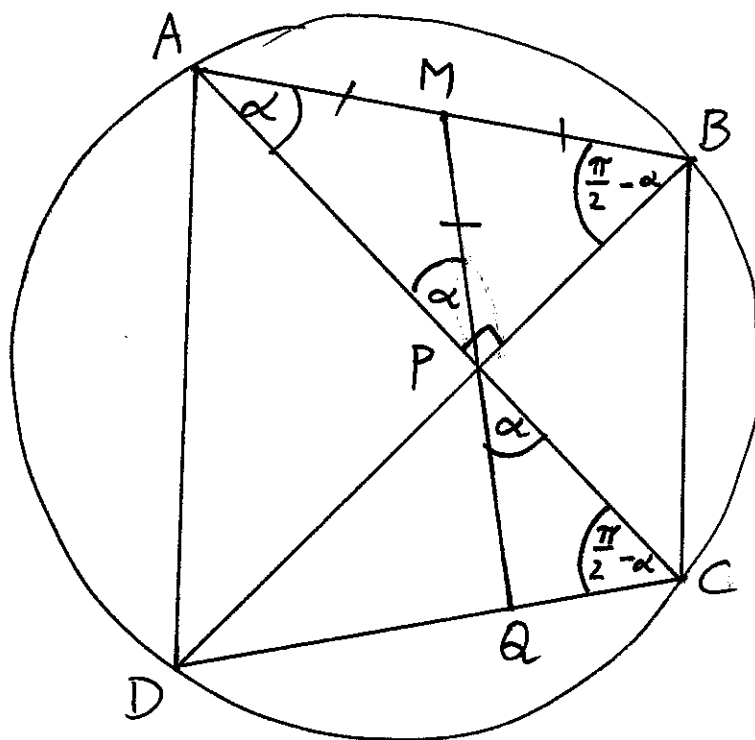
$x = 2$

$y = x$



(15)(a)

(9)



(i) $\angle APB = \frac{\pi}{2}$, so it is an angle in a semicircle with diameter AB.

So M is the centre of the semicircle.

So AM and PM are equal radii. ✓

(ii) $AM = PM$ (from (i)),

so $\angle APM = \alpha$ (base angles of isosceles triangle), } ✓

so $\angle CPQ = \alpha$ (vertically opposite).

Now, $\angle ABD = \frac{\pi}{2} - \alpha$ (angle sum of $\triangle APB$), ✓

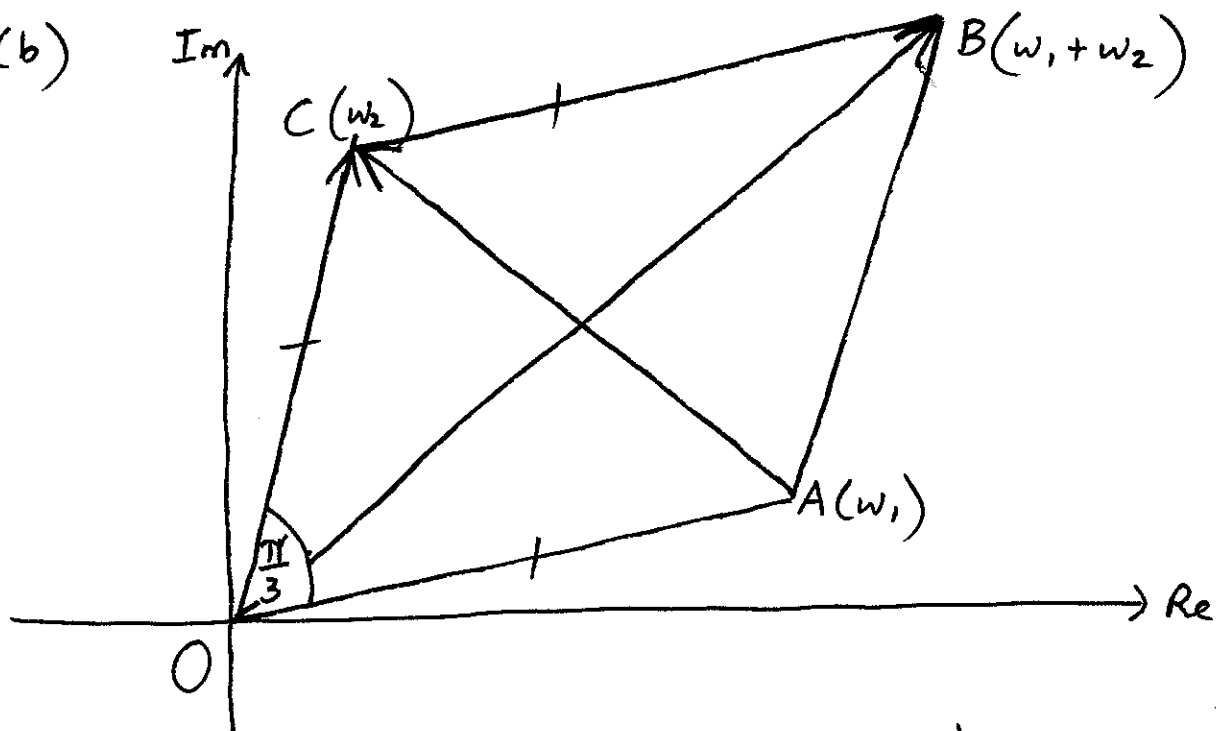
so $\angle ACD = \frac{\pi}{2} - \alpha$ (angles at circumference on same arc) } ✓

so $\angle MQC = \frac{\pi}{2}$ (angle sum of $\triangle PQC$). } ✓

So $MQ \perp DC$.

(15)(b)

(10)



(i) $\vec{OC}$ is the anticlockwise rotation of $\vec{OA}$ through $\frac{\pi}{3}$ about O. So $w_2 = w_1 \cdot 1 \operatorname{cis} \frac{\pi}{3}$
 $= w_1 \operatorname{cis} \frac{\pi}{3}$.

(ii) $\vec{OB}$ represents $w_1 + w_2 = w_1 + w_1 \operatorname{cis} \frac{\pi}{3}$
 $= w_1 (1 + \operatorname{cis} \frac{\pi}{3})$

and $\vec{AC}$ represents $w_2 - w_1 = w_1 \operatorname{cis} \frac{\pi}{3} - w_1$
 $= w_1 (\operatorname{cis} \frac{\pi}{3} - 1)$.

$$\begin{aligned}
 \text{(iii)} \quad i(w_1 + w_2) &= iw_1 (1 + \operatorname{cis} \frac{\pi}{3}) \\
 &= iw_1 \left(\frac{3}{2} + \frac{\sqrt{3}}{2} i \right) \\
 &= w_1 \left(-\frac{\sqrt{3}}{2} + \frac{3}{2} i \right) \\
 &= \sqrt{3} w_1 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2} i \right) \\
 &= \sqrt{3} w_1 (\operatorname{cis} \frac{\pi}{3} - 1) \\
 &= \sqrt{3} (w_2 - w_1)
 \end{aligned}$$

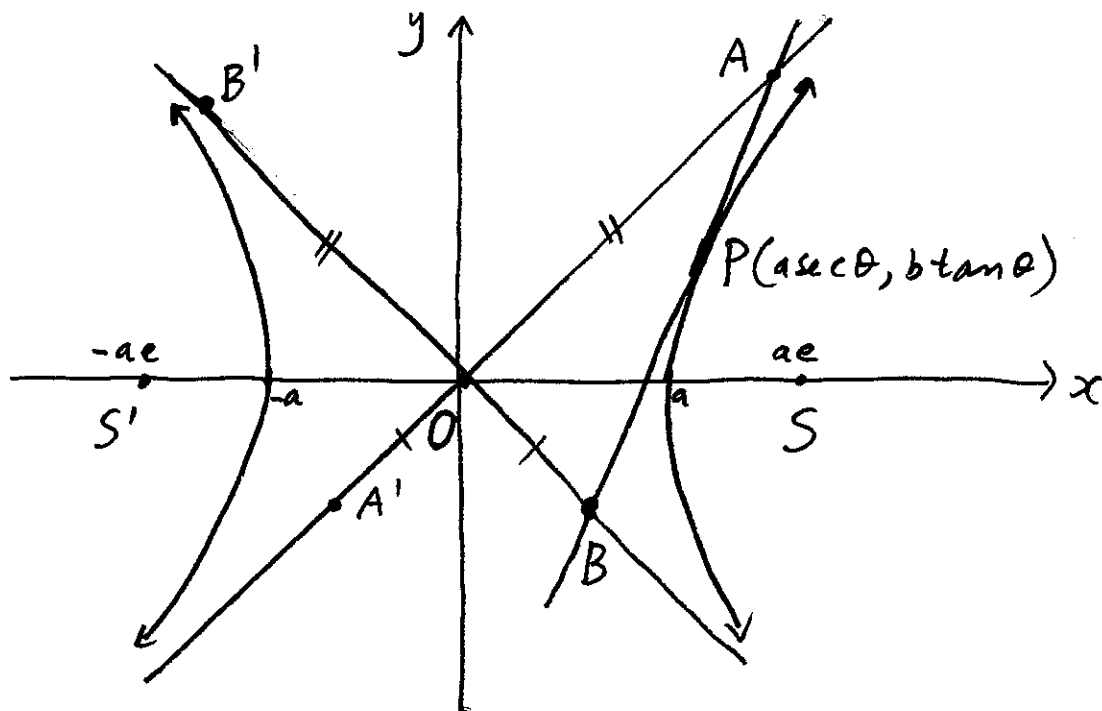
$$\text{So } w_2 - w_1 = \frac{1}{\sqrt{3}} i (w_1 + w_2).$$

$$\text{So } w_2 - w_1 = (w_1 + w_2) \cdot r \operatorname{cis} \frac{\pi}{2}, \text{ where } r = \frac{1}{\sqrt{3}}.$$

So AC and OB are perpendicular.

(15)(c)

(11)



$$(i) \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{b \sec^2 \theta}{a \sec \theta \tan \theta} = \frac{b \sec \theta}{a \tan \theta}$$

So the tangent at P has equation

$$y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta) \quad \checkmark$$

$$ay \tan \theta - ab \tan^2 \theta = bx \sec \theta - ab \sec^2 \theta$$

$$bx \sec \theta - ay \tan \theta = ab$$

(ii) Solve the tangent equation simultaneously with $y = \frac{bx}{a}$:

$$bx \sec \theta - bx \tan \theta = ab$$

$$x = \frac{a}{\sec \theta - \tan \theta} \cdot \frac{\sec \theta + \tan \theta}{\sec \theta + \tan \theta} = a(\sec \theta + \tan \theta)$$

$$\text{So } A = (a(\sec \theta + \tan \theta), b(\sec \theta + \tan \theta)) \quad \checkmark$$

$$\text{Similarly, } B = (a(\sec \theta - \tan \theta), b(\sec \theta - \tan \theta)) \quad \checkmark$$

$$\text{So } OA^2 \cdot OB^2 = (a^2(\sec \theta + \tan \theta)^2 + b^2(\sec \theta + \tan \theta)^2) \cdot (a^2(\sec \theta - \tan \theta)^2 + b^2(\sec \theta - \tan \theta)^2)$$

$$= (a^2 + b^2)(\sec \theta + \tan \theta)^2 \cdot (a^2 + b^2)(\sec \theta - \tan \theta)^2 \quad \checkmark$$

$$= (a^2 + b^2)^2 (\sec^2 \theta - \tan^2 \theta)^2$$

$$\text{So } OA \cdot OB = a^2 + b^2$$

(15)(c)(iii) $OA \cdot OA' = OB \cdot OB'$, by the definitions of A' and B' .

So, by the converse of the intersecting chords theorem, the points A, B, A' and B' are concyclic. ✓

$$(iv) OA \cdot OA' = OB \cdot OB'$$

$$= OA \cdot OB$$

$$= a^2 + b^2 \quad (\text{from (ii)}) \quad \checkmark$$

$$\text{But } OS \cdot OS' = a^2 e^2$$

$$= a^2 + b^2 \quad (\text{since } b^2 = a^2(e^2 - 1)) \quad \checkmark$$

So, by the converse of the intersecting chords theorem, S and S' lie on the same circle as A, B, A' and B' .

So A, B, S and S' are concyclic.

$$\begin{aligned}
 (16)(a)(i) \quad z^n + z^{-n} &= (\operatorname{cis} \theta)^n + (\operatorname{cis} \theta)^{-n} \\
 &= \operatorname{cis} n\theta + \operatorname{cis}(-n\theta) \quad (\text{de Moivre}) \\
 &= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta \\
 &= 2 \cos n\theta
 \end{aligned}$$

$$\begin{aligned}
 (ii) \text{ LHS} &= (z^{2n} + z^{-2n}) + (z^{2n-2} + z^{-(2n-2)}) + \dots + (z^2 + z^{-2}) + z^0 \sin \theta \\
 &= (2 \cos 2n\theta + 2 \cos(2n-2)\theta + \dots + 2 \cos 2\theta + 1) \sin \theta \\
 &= (\sin(2n+1)\theta - \sin(2n-1)\theta) + (\sin(2n-1)\theta - \sin(2n-3)\theta) \\
 &\quad + \dots + (\sin 3\theta - \sin \theta) + \sin \theta \\
 &= \sin(2n+1)\theta = \text{RHS}
 \end{aligned}$$

(iii) Let $n=3$.

$$\text{Then } (2 \cos 6\theta + 2 \cos 4\theta + 2 \cos 2\theta + 1) \sin \theta = \sin 7\theta$$

$$\begin{aligned}
 2(4 \cos^3 2\theta - 3 \cos 2\theta) + 2(2 \cos^2 2\theta - 1) + 2 \cos 2\theta + 1 &= \frac{\sin 7\theta}{\sin \theta} \\
 8 \cos^3 2\theta + 4 \cos^2 2\theta - 4 \cos 2\theta - 1 &= \frac{\sin 7\theta}{\sin \theta}
 \end{aligned}$$

(iv) Let $\theta = \frac{\pi}{7}$.

$$\text{Then } 8 \cos^3 \frac{2\pi}{7} + 4 \cos^2 \frac{2\pi}{7} - 4 \cos \frac{2\pi}{7} - 1 = \frac{\sin \pi}{\sin \frac{\pi}{7}} = 0.$$

So it follows that $\cos \frac{2\pi}{7}$ is a root of the equation

$$8x^3 + 4x^2 - 4x - 1 = 0.$$

$$(16)(b)(i) \quad \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

14

$$\text{Let } \theta = \frac{\pi}{8}.$$

$$\text{Then } \tan \frac{\pi}{4} = \frac{2t}{1-t^2}, \text{ where } t = \tan \frac{\pi}{8}$$

$$t^2 + 2t - 1 = 0$$

$$(t+1)^2 = 2$$

$$t = \pm\sqrt{2} - 1$$

$$\text{But } t > 0, \text{ so } t = \tan \frac{\pi}{8} = \sqrt{2} - 1.$$

$$(ii) \quad \cos 4\theta = 2\cos^2 2\theta - 1$$

$$= 2(2\cos^2 \theta - 1)^2 - 1$$

$$= 2(4\cos^4 \theta - 4\cos^2 \theta + 1) - 1$$

$$= 8\cos^4 \theta - 8\cos^2 \theta + 1$$

$$(iii)(a)$$

$$I = \int_{-\frac{\pi}{8}}^{\frac{\pi}{8}} \frac{4\cos 4\theta}{\sqrt{1+\sin 4\theta} + \sqrt{1-\sin 4\theta} + 2} d\theta$$

$$= \int_{-\frac{\pi}{8}}^{\frac{\pi}{8}} \frac{4\cos 4\theta}{\sqrt{\cos^2 2\theta + \sin^2 2\theta + 2\sin 2\theta \cos 2\theta} + \sqrt{\cos^2 2\theta + \sin^2 2\theta - 2\sin 2\theta \cos 2\theta} + 2} d\theta$$

$$x = \sin 4\theta$$

$$dx = 4\cos 4\theta d\theta$$

x	-1	1
θ	$-\frac{\pi}{8}$	$\frac{\pi}{8}$

$$= \int_{-\frac{\pi}{8}}^{\frac{\pi}{8}} \frac{4\cos 4\theta}{(\cos 2\theta + \sin 2\theta) + (\cos 2\theta - \sin 2\theta) + 2} d\theta$$

$$= 2 \int_0^{\frac{\pi}{8}} \frac{4\cos 4\theta}{2(\cos 2\theta + 1)} d\theta$$

$$= 2 \int_0^{\frac{\pi}{8}} \frac{2\cos 4\theta}{2\cos^2 \theta} d\theta$$

$$= \int_0^{\frac{\pi}{8}} \frac{2\cos 4\theta}{\cos^2 \theta} d\theta$$

$$(B) \quad I = 2 \int_0^{\frac{\pi}{8}} \frac{8\cos^4 \theta - 8\cos^2 \theta + 1}{\cos^2 \theta} d\theta$$

$$= 2 \int_0^{\frac{\pi}{8}} (8\cos^2 \theta - 8 + \sec^2 \theta) d\theta$$

$$= 2 \int_0^{\frac{\pi}{8}} (4 + 4\cos 2\theta - 8 + \sec^2 \theta) d\theta$$

$$= 2 [2\sin 2\theta - 4\theta + \tan \theta]_0^{\frac{\pi}{8}}$$

$$= 2 \left(2\sqrt{2} - \frac{\pi}{2} + (\sqrt{2} - 1) \right)$$

$$= 4\sqrt{2} - \pi - 2$$

(16)(c)(i) LHS = $\frac{1}{1 \times 2} + \frac{1}{3 \times 4} + \frac{1}{5 \times 6} + \dots + \frac{1}{(2k-1)(2k)}$ (15)

$$> \frac{1}{(2k+1)(2k+2)} + \frac{1}{(2k+1)(2k+2)} + \dots + \frac{1}{(2k+1)(2k+2)} \quad (k \text{ terms})$$

$$= \frac{k}{(2k+1)(2k+2)}$$

$$= \text{RHS}$$

(ii) Proof by induction:

When $n=2$, LHS = $2 \times \frac{4}{3} = 2\frac{2}{3}$ and RHS = $3 \times \frac{3}{4} = 2\frac{1}{4}$.

So the result is true when $n=2$.

Assume that the result is true for the positive integer $n=k$.

That is, assume that $k(1 + \frac{1}{3} + \dots + \frac{1}{2k-1}) > (k+1)(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2k})$. (*)

Prove that the result is true for $n=k+1$.

That is, prove that

$$(k+1)(1 + \frac{1}{3} + \dots + \frac{1}{2k-1} + \frac{1}{2k+1}) > (k+2)(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2k} + \frac{1}{2k+2}).$$

$$\text{LHS} = (k+1)(1 + \frac{1}{3} + \dots + \frac{1}{2k-1} + \frac{1}{2k+1})$$

$$= k(1 + \frac{1}{3} + \dots + \frac{1}{2k-1}) + 1(1 + \frac{1}{3} + \dots + \frac{1}{2k-1}) + \frac{k+1}{2k+1}$$

$$> (k+1)(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2k}) + (\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2k} + \frac{k}{(2k+1)(2k+2)}) + \frac{k+1}{2k+1}$$

(using *)

$$= (k+2)(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2k}) + \frac{k + (k+1)(2k+2)}{(2k+1)(2k+2)}$$

(using part (i))

$$= (k+2)(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2k}) + \frac{(2k+1)(k+2)}{(2k+1)(2k+2)}$$

$$= \text{RHS}$$

So, by mathematical induction, the result is true for all $n \geq 2$.

(16)(c)(ii) Otherwise:

$$\begin{aligned}
 1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2n-1} &> 1 + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2n} \\
 &= \frac{1}{2} + \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2n} \right) \\
 &> \frac{1}{n} \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2n} \right) \\
 &\quad + \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2n} \right),
 \end{aligned}$$

$$\begin{aligned}
 \text{since } \frac{n}{2} &= \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots + \frac{1}{2} \quad (n \text{ terms}) \\
 &> \frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2n}
 \end{aligned}$$

$$\text{so that } \frac{1}{2} > \frac{1}{n} \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2n} \right).$$

$$\begin{aligned}
 \text{So } 1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2n-1} &> \left(\frac{1}{n} + 1 \right) \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2n} \right) \\
 &= \frac{n+1}{n} \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2n} \right),
 \end{aligned}$$

$$\text{so } n \left(1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2n-1} \right) > (n+1) \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2n} \right).$$